The small diagonal suffixes in the equality case

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses, the actual intersection property and all intransitive ambient omissions. Use the explicit two-prefix-orbit equality interface at a unique interior cut $j = g + 1$:

$$P = P_{g-1}, \quad \mathrm{supp}(P) = K \dot\cup E, \quad |K| + |E| = 2g + 1, \quad \mathrm{Fix}(P) = A \cup \{b\}.$$

Here K is the head orbit, E is one exterior orbit, $C = C_g$ is the head-side omission orbit and $A = \Omega \setminus C$. Every label $3, \dots, g-1$ is paired in the full ambient sense. The unique crossing of $s = \rho_{g+1}$ is $(b\ a_0)$. These hypotheses are explicit; this note does not infer them for other cuts or other numbers of prefix orbits.

Lemma 1 (The crossing fixes the old prefix support)**.** *The generator s fixes $K \cup E$ pointwise. Consequently it has the form $s = (b\ a_0)w$, where w is an odd involution on $A \setminus \{a_0\}$.*

Proof. Far commutation makes s centralize P. The pure alternating kernel on K forces it to fix K pointwise, and it preserves the only other nontrivial P-orbit E. Thus $s = (b\ a_0)fw$ with disjoint factors on E and $A \setminus \{a_0\}$. Suppose $f \neq 1$. A nonidentity element centralizing a transitive permutation group cannot fix a point, so f is a fixed-point-free involution on E. Write $|E| = 2m$.

The exact actual prefix intersection identifies the two nontrivial orbits of $H = G_{3,\dots,g-1}$ as $T = K \setminus (D \setminus \{6\})$ and E, where $|T| = |K| - 5$. Let L consist of the labels in $I = \{3, \dots, g-1\}$ having an ambient-eligible edge in E, and put $F = I \setminus L$. No eligible edge can be one of the pairs of f, since $s \in G_i$ for every $i \in I$.

Select one eligible edge in E for each label in L and project to the m pairs of f. These selected edges form a forest. A cycle would supply, for one selected i-edge, an avoiding path between its two pairs. All the generators in H commute with f; the path lifts to E, and s supplies any required switches within a pair. This gives an actual G_i path between the endpoints of an eligible i-edge, a contradiction. Hence $|L| \leq m - 1$.

Every eligible edge for a label in F lies in T, and each such label is paired. The acquired selected-localization theorem, applied to the arbitrary subset F of the interval I and its invariant orbit T, gives $2|F| \leq |T| - 1 = |K| - 6$. Therefore

$$g - 3 = |L| + |F| \leq \frac{|E|}{2} - 1 + \frac{|K| - 6}{2} = \frac{2g - 7}{2} < g - 3.$$

Thus $f = 1$. Since s is even, w is odd. No sign assumption on f was used in the contradiction; this strengthens the preceding exterior-deficit statement. □

For the remaining statements assume that the diagonal suffix has the actual marking supplied by the equality path argument:

$$A = \{a_0, a_0', \dots, a_{t-1}, a_{t-1}'\}, \qquad q_i = \rho_{g+1+i} = (a_{i-1}\ a_i)(a_{i-1}'\ a_i') \quad (1 \leq i < t).$$

Here $t = r - g - 1$, and $u = \rho_g$ acts on A as $v = \prod_{i=0}^{t-1}(a_i\ a_i')$. The actual prefix P_g is transitive on C. It follows that u moves b: otherwise every generator of P_g would fix b. Write $c = b^u \in C \setminus \{b\}$.

Theorem 2 (The six-point suffix is impossible)**.** *The above equality case cannot have $t = 3$.*

Proof. The generator s centralizes $q_2 = (a_1\ a_2)(a_1'\ a_2')$. Its unique crossing sends b to a_0; since it fixes $C \setminus \{b\}$ by Lemma 1, it must fix the only remaining fixed point a_0' of q_2. Its action on the four other points is an odd involution centralizing the displayed double transposition. Such an involution is exactly one of $(a_1\ a_2)$ and $(a_1'\ a_2')$: a commuting permutation either preserves both two-point orbits, in which case this is the odd part of their direct product, or exchanges them, in which case an involution is a product of two transpositions and is even. Hence

$$s = (b\ a_0)(a_1\ a_2) \quad \text{or} \quad s = (b\ a_0)(a_1'\ a_2').$$

Consider the actual omission of the last mark q_2. It contains P_g, so C is connected in its orbit graph. The crossing joins C to a_0, and the three edges of $u|_A$ join each a_i to a_i'. The edges of q_1 join the pair with index 0 to the pair with index 1. In either displayed form of s, its remaining edge joins the pair with index 1 to the pair with index 2. Thus this omission is transitive on $C \cup A = \Omega$, contrary to hypothesis. This is an actual orbit argument; no intersection property is asserted for a projected marking. □

Theorem 3 (The four-point suffix reduces to one form)**.** *If $t = 2$, then necessarily*

$$q_1 = (a_0\ a_1)(a_0'\ a_1'), \qquad s = (b\ a_0)(a_1\ a_1'), \qquad u = (b\ c)(a_0\ a_0')(a_1\ a_1')\delta,$$

where δ is an odd involution supported on $C \setminus \{b, c\}$. This statement is a reduction, not an exclusion of the displayed form.

Proof. By Lemma 1, the restriction of s to $\{a_0', a_1, a_1'\}$ is an odd involution, hence a single transposition. If it is $(a_0'\ a_1)$ or $(a_0'\ a_1')$, the crossing, the two pair edges of u and this transposition join every point of A to C without using q_1. Together with the transitive P_g action on C, they make the last ambient omission transitive. Only $(a_1\ a_1')$ remains. The involution u preserves A and C, has the stated action on A and exchanges b, c. Its other factor is supported on $C \setminus \{b, c\}$. The three displayed transpositions are odd, so global evenness forces δ to be odd. □

Attribution and scope. The first lemma strengthens the exterior-deficit argument using the same acquired selected-localization theorem and its full predecessors. The six-point omission proof is the orbit argument already used in the accepted small anchored-imprimitive proof, applied here inside the actual ambient group. The literal four-point reduction records the remaining decorated dihedral case; its exterior factor has not been removed. The submitted diagonal row-swap theorem separately excludes the large remaining centralizer action for $t \geq 4$. Other prefix-orbit cases, endpoint cuts, opposite heads and terminal singleton remain. No new computation, formal theorem, whole-original or question credit is claimed.